

# XDR vs SOAR

Detection, Orchestration, and Response  
in Cloud-Centric Advanced Monitoring Architectures

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*A research document examining how XDR and SOAR differ, what solutions exist for each, how they are implemented, and what changes when the target environment is built primarily on cloud infrastructure.*

## **Abstract**

Extended Detection and Response (XDR) and Security Orchestration, Automation, and Response (SOAR) both aim to reduce the time between threat detection and containment, but they attack the problem from opposite ends. XDR owns the full pipeline from telemetry collection through cross-layer correlation to automated response. SOAR sits downstream, accepting alerts from other tools and automating the multi-step workflows that follow. This document breaks down what each technology actually does, compares their architectures side by side, surveys the major commercial and open-source solutions in both categories, and walks through how each is implemented in practice. A dedicated section examines what changes when the target environment is built on cloud infrastructure: ephemeral workloads, multi-cloud sprawl, identity-first attack surfaces, and the cost implications of high-volume cloud telemetry. The document concludes with a practical assessment of when to use each, when to use both, and what the deciding factors are for organisations designing an advanced monitoring system.

**Keywords:** XDR, SOAR, Extended Detection and Response, Security Orchestration, Cloud Security, SIEM, Advanced Monitoring, Incident Response, MITRE ATT&CK, Threat Detection, Playbooks

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## 1. Introduction

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Security teams in any mid-to-large organisation now manage more tools than they can reasonably operate. The average enterprise security stack includes dozens of products—each generating its own alerts, storing its own logs, and exposing its own management interface [1]. Analysts spend more time switching between consoles and manually stitching together events than they do actually investigating threats.

Two categories of platform have emerged to address this: Extended Detection and Response (XDR) and Security Orchestration, Automation, and Response (SOAR). Both promise to speed up detection and response. Both promise to reduce manual work. But they solve different parts of the problem, and choosing between them—or deciding to use both—requires understanding what each actually does under the hood.

This document is structured as follows. Sections 2 and 3 define each technology in detail: architecture, internal mechanics, data flows, and limitations. Section 4 compares them directly across every relevant dimension. Sections 5 and 6 survey the solution landscape for each. Section 7 covers practical implementation. Section 8 focuses specifically on cloud environments. Section 9 discusses combined deployments. Section 10 wraps up with practical guidance.

## 2. What XDR Is and How It Works

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Extended Detection and Response is a security platform that collects telemetry from multiple layers of an organisation’s environment—endpoints, the network, cloud services, identity systems, and email—and correlates it into unified incidents for detection and response. It replaces the model where each layer has its own tool and its own alert queue with a single platform that sees everything at once [3].

### 2.1 Architecture

An XDR platform has four main components.

The **data ingestion layer** collects raw telemetry. On endpoints, this means process execution, file system activity, registry changes, and memory behaviour, typically gathered by a lightweight agent. On the network, it means flow records, DNS queries, proxy logs, and firewall events. From cloud providers, it means API audit trails, configuration change logs, and workload telemetry. From identity systems, it means authentication events, privilege changes, token issuance, and session metadata. All of this data is normalised into a common schema so the platform can compare and join events from different sources [4].

The **detection engine** applies multiple detection methods in parallel. Rule-based detec-

tion matches activity against signatures of known threats—fast and low on false positives, but blind to anything that has not been catalogued yet. Behavioural detection builds a baseline of what normal looks like for each user and device and flags deviations from it—better at catching insider threats and novel techniques, but slower to converge and prone to false positives during its learning period. Machine learning models classify event sequences at a scale that neither rules nor baselines can match manually: supervised models handle known attack classes, unsupervised models flag statistical anomalies [17]. Threat intelligence lookups check indicators (IP addresses, domains, file hashes) against feeds of known malicious infrastructure [16].

The **correlation engine** is what separates XDR from a stack of point tools. It links related events from different data sources into a single incident. A suspicious PowerShell execution on an endpoint, an anomalous DNS query from the same host, and a login attempt from an unfamiliar geolocation using the same account are not three separate alerts—they are one attack chain. The correlation engine assembles the chain, assigns a severity score, maps the activity to MITRE ATT&CK techniques, and presents it as a single incident with a timeline [3].

The **response engine** can act on what the detection and correlation engines find. It can isolate an endpoint from the network, kill a process, block a domain or IP, disable a user account, quarantine an email, or revoke a cloud session—all without waiting for a human to click a button. Which actions run automatically and which wait for analyst approval is configurable per detection type, confidence level, and asset criticality.

## 2.2 Native XDR vs Open XDR

**Native XDR** means every component—agent, sensor, cloud connector, detection logic—comes from a single vendor. Data formats are consistent, integration is seamless, and detection depth is high. The price is vendor lock-in. Switching away means migrating detections, workflows, and institutional knowledge [1].

**Open XDR** provides the correlation and detection layer on top of existing tools, ingesting data through APIs and standard formats. This preserves current investments and avoids single-vendor dependency. The trade-off is that detection quality depends on what each source tool exposes through its API, and integration effort is higher.

Most production deployments sit between the two: one vendor’s platform as the core, with a handful of third-party feeds connected where needed.

### 2.3 What XDR Does Not Do

XDR is not a ticketing system. Its case management is typically limited to incident tracking within the platform. It does not manage the full response lifecycle the way a dedicated case management tool does.

XDR is not a general automation platform. Its response actions are deep within its own ecosystem but narrow outside it. If an organisation needs to trigger actions on tools the XDR vendor does not integrate with—a niche firewall, a custom internal system, a particular cloud service—the XDR platform usually cannot reach them.

XDR does not replace a SIEM for compliance-driven long-term log retention. Most XDR platforms retain detailed telemetry for 30–180 days. Regulatory requirements that demand years of retention still need a separate archival pipeline.

## 3. What SOAR Is and How It Works

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Security Orchestration, Automation, and Response is a platform that connects security tools together and automates the multi-step workflows that run when an incident is detected. SOAR does not collect telemetry and does not perform detection. It takes alerts generated by other systems—SIEM, EDR, cloud security tools, email gateways—and processes them through predefined or analyst-triggered workflows [2].

### 3.1 The Three Functions

**Orchestration** is the ability to talk to other tools. A SOAR platform integrates with firewalls, endpoint agents, SIEM, identity providers, ticketing systems, cloud platforms, threat intelligence services, DNS providers, email platforms, and anything else that exposes an API. Each integration registers a set of actions: “block this IP”, “isolate this host”, “look up this hash”, “create this ticket”. The SOAR platform can call any of these actions from within a workflow [7].

**Automation** is the execution of playbooks—predefined sequences of steps that run when a specific type of alert arrives. A phishing playbook might extract URLs and attachments from the reported email, submit attachments to a sandbox, query reputation services for the URLs, search the SIEM for other users who received the same message, check proxy logs for anyone who clicked the link, isolate any affected endpoints, force password resets for any affected accounts, remove the email from all inboxes, and create an incident ticket. All of this can happen in seconds, with full logging of every step [7].

**Case management** provides a structured record of every incident from first alert through investigation, containment, eradication, and recovery. Analysts collaborate on cases, attach evidence, log decisions, and track whether all required response steps have been

completed. This is the layer that produces the audit trail compliance teams and legal counsel ask for [18].

### 3.2 Playbook Anatomy

A playbook is a directed graph. Each node is one of three types:

An **action node** calls an integration: enrich an indicator, query a database, block a hash, send a notification. Actions execute via the integration’s API and return structured output that downstream nodes can use.

A **decision node** branches the workflow based on a condition: if the threat score is above 80, take the automated containment path; if below 80, route to an analyst for review.

A **human task node** pauses execution and waits for an analyst to perform a step or approve an action. This is the mechanism for keeping a human in the loop when the risk of a wrong automated action is too high.

Production playbooks are rarely linear. They include parallel branches (sandbox the file and query the reputation service at the same time), error handling (what to do when the sandbox API times out), retry logic, loops (iterate over every recipient of a phishing email), and sub-playbooks (call a shared “enrich indicator” playbook from multiple parent playbooks).

### 3.3 What SOAR Does Not Do

SOAR does not generate detections. It does not watch endpoints, networks, or cloud APIs for suspicious activity. It waits for someone else to tell it that something happened, then decides what to do about it.

This means SOAR’s effectiveness has a hard ceiling set by the quality of its inputs. If the upstream SIEM fires thousands of low-quality alerts with minimal context, the SOAR platform will process thousands of low-quality alerts with minimal context. Automating bad input does not make it good—it makes it faster.

SOAR also does not collect or store telemetry. It has no historical data to query for threat hunting. It cannot correlate events across sources the way XDR does because it never sees the raw events—only the alerts that other tools have already generated from them.

## 4. XDR vs SOAR — Direct Comparison

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## 4.1 Architecture

XDR is a vertically integrated platform. It owns the full stack: data collection, normalisation, detection, correlation, and response. SOAR is a horizontal integration layer. It sits on top of other tools and connects them through APIs, without owning any part of the detection pipeline.

This architectural difference drives every other difference between the two.

## 4.2 Comparison Matrix

Table 1: XDR vs SOAR across key dimensions

| Dimension                | XDR                                                             | SOAR                                                            |
|--------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Core purpose</b>      | Unified detection and response across security layers           | Workflow automation and cross-tool orchestration                |
| <b>Telemetry</b>         | Collects raw data via agents, sensors, and cloud APIs           | Does not collect telemetry; receives alerts from other tools    |
| <b>Detection</b>         | Built-in engine: rules, ML, behavioural baselines, threat intel | None—entirely dependent on upstream detection tools             |
| <b>Correlation</b>       | Automatic cross-layer correlation into incidents                | Alert enrichment via API lookups; no raw event correlation      |
| <b>Response scope</b>    | Deep within its own ecosystem; limited outside it               | Broad across any tool with an API integration                   |
| <b>Playbooks</b>         | Built-in, moderate complexity, tied to detection types          | Advanced builder: branching, loops, human tasks, error handling |
| <b>Case management</b>   | Basic incident tracking                                         | Full lifecycle with collaboration, evidence, audit trail        |
| <b>Integration count</b> | Tens (own ecosystem plus select third-party)                    | Hundreds (any tool with an API)                                 |
| <b>Deployment</b>        | Agent-based plus cloud connectors                               | API-based; no agents deployed                                   |
| <b>Cloud support</b>     | Varies by vendor; deepest with native cloud provider            | Inherently multi-cloud via per-provider API integrations        |
| <b>Best for</b>          | Replacing fragmented detection with unified visibility          | Automating and standardising existing response workflows        |

### 4.3 Detection Quality vs Response Breadth

This is the core trade-off. XDR improves detection quality—it reduces the volume of false positives analysts see because its correlation engine filters weak signals before they become incidents. SOAR does not reduce false positives. It processes whatever it receives, including every noisy alert the upstream SIEM produces.

In the other direction, SOAR’s response reach is wider. Because it integrates with hundreds of tools via APIs, it can trigger actions on systems that the XDR platform has no visibility into. XDR’s response is fast and deep but limited to what it manages. SOAR’s response can touch anything with an API but depends on the reliability and latency of each integration.

### 4.4 When to Choose One Over the Other

Choose XDR when the primary problem is too many disconnected detection tools producing too many uncorrelated alerts. XDR directly addresses alert fatigue and blind spots between tools.

Choose SOAR when detection is already solid—a mature SIEM deployment or a strong EDR product—but response is slow, inconsistent, and manual. SOAR directly addresses the gap between “we detected it” and “we contained it”.

Choose both when the organisation is large enough that detection quality and response automation are both bottlenecks. XDR feeds high-quality correlated incidents to SOAR, and SOAR orchestrates the extended response actions that stretch beyond XDR’s ecosystem.

## 5. XDR Solutions

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### 5.1 Commercial Platforms

#### 5.1.1 Microsoft Defender XDR

Built into the Microsoft 365 security stack. Correlates signals from Defender for Endpoint, Defender for Office 365, Defender for Identity, Defender for Cloud Apps, and Microsoft Sentinel. Advanced hunting uses Kusto Query Language (KQL) across all data sources, so a single query can join a process execution on an endpoint with a sign-in event in Entra ID and a mail delivery event in Exchange Online [4].

Cloud coverage is strongest for Azure: Defender for Cloud provides workload protection across Azure VMs, AKS, SQL, and storage, with alerts feeding straight into the XDR incident queue. AWS and GCP connectors exist but offer less depth.

Automated Investigation and Response (AIR) handles common incident types without analyst intervention. For anything more complex, Microsoft Sentinel’s SOAR capabilities (Logic Apps-based playbooks) extend the automation layer.

**Strengths:** deepest native integration for Microsoft-heavy environments; KQL is expressive and fast; licensing is bundled for E5 customers.

**Weaknesses:** significantly less useful outside the Microsoft ecosystem; Sentinel costs scale with data volume; the number of consoles (Defender portal, Sentinel, Azure portal, Entra) is still higher than it should be.

### 5.1.2 CrowdStrike Falcon

Cloud-native platform that started in EDR and expanded into XDR through Falcon Identity Threat Detection, Falcon Cloud Security, Falcon for Mobile, and Falcon Intelligence. The Threat Graph processes trillions of events weekly, mapping relationships between endpoints, identities, and infrastructure [6].

The lightweight agent is a deployment strength: low resource usage on endpoints, fast rollout, and broad OS coverage including Linux. For cloud environments, Falcon Cloud Security covers CSPM (Cloud Security Posture Management) and CWP (Cloud Workload Protection) across AWS, Azure, and GCP, with container runtime visibility and Kubernetes audit log ingestion.

**Strengths:** fast detection; excellent threat intelligence; lightweight agent; strong multi-cloud support.

**Weaknesses:** full value requires multiple Falcon modules; third-party correlation depth is shallower than native; premium pricing.

### 5.1.3 Palo Alto Networks Cortex XDR

One of the first platforms to carry the XDR label. Integrates endpoint, network, and cloud telemetry using the Cortex agent, Palo Alto firewalls (via Cortex Data Lake), and Prisma Cloud. The behavioural analytics engine baselines user and device activity. XQL (XDR Query Language) provides cross-source hunting [1].

Network integration is the differentiator: organisations running Palo Alto firewalls get deep network telemetry—URL filtering events, WildFire sandbox results, GlobalProtect VPN data—without additional connectors.

**Strengths:** strong network integration; mature behavioural analytics; Prisma Cloud adds cloud posture and workload protection.

**Weaknesses:** best value is locked to Palo Alto firewall customers; Cortex Data Lake

costs scale with ingestion volume.

#### 5.1.4 Elastic Security

Built on the Elastic Stack (Elasticsearch, Kibana, Elastic Agent). All detection rules are open-source, mapped to ATT&CK, and maintained by Elastic’s threat research team alongside community contributors. Available as a managed cloud service (Elastic Cloud) or self-hosted [10].

Cloud integrations cover AWS (CloudTrail, GuardDuty, VPC Flow Logs), Azure (Activity Logs, Defender), and GCP (Cloud Audit Logs, Security Command Centre). The open approach treats all data sources equally, which is an advantage in multi-cloud environments where no single vendor dominates.

**Strengths:** open detection rules; flexible data ingestion; self-hostable; no vendor lock-in on detection logic.

**Weaknesses:** self-hosted deployments require significant operational effort; agent maturity trails CrowdStrike and Microsoft; response capabilities are less automated out of the box.

## 5.2 Open-Source Options

No single open-source project delivers full XDR. The closest approximation is a stack:

**Wazuh** provides host-based intrusion detection, file integrity monitoring, vulnerability scanning, and log analysis through a lightweight agent. Cloud modules cover AWS CloudTrail, VPC Flow Logs, Azure, and GCP. Wazuh is the backbone for endpoint and log visibility in most open-source security stacks [11].

**Suricata + Zeek** together cover network detection. Suricata handles signature-based detection and IPS functionality. Zeek generates rich protocol metadata and connection logs. Running both gives the stack both signature hits and deep traffic analysis.

**Velociraptor** adds real-time endpoint interrogation and forensic artifact collection using VQL. It excels at fleet-wide hunting—querying thousands of endpoints in parallel for specific indicators or configurations [12].

**OpenSearch Security Analytics** provides the SIEM-like correlation layer, with detection rules and dashboards.

The integration work is the main cost. Normalising data from Wazuh, Suricata, Zeek, and Velociraptor into a common schema, building cross-source correlation rules, and wiring up response actions requires engineering effort that commercial platforms handle out of the box.

## 6. SOAR Solutions

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### 6.1 Commercial Platforms

#### 6.1.1 Palo Alto Networks Cortex XSOAR

Formerly Demisto. Visual playbook builder with branching, loops, conditional logic, human-in-the-loop approval, and error handling. The integration marketplace has over 800 content packs covering firewalls, EDR, SIEM, ticketing, cloud platforms, and threat intelligence [7].

The War Room provides per-incident collaboration: analysts run commands, share findings, and log decisions in a single space. The built-in Threat Intelligence Management (TIM) module aggregates feeds, deduplicates indicators, and scores them.

Cloud integrations cover AWS (IAM, Security Hub, GuardDuty, S3, EC2, Lambda), Azure (Sentinel, Defender, Entra ID), and GCP (Security Command Centre, IAM, Compute). A single playbook can orchestrate cross-cloud actions: revoke an AWS access key, block an IP in an Azure NSG, and open a Jira ticket, all in one sequence.

**Strengths:** broadest integration library; strong playbook builder; active marketplace.

**Weaknesses:** expensive at scale; playbook complexity can grow out of control without governance; steep learning curve for advanced features.

#### 6.1.2 Splunk SOAR (formerly Phantom)

Tight integration with Splunk Enterprise and Splunk Cloud. Playbooks use a visual editor or direct Python for complex logic. Over 350 integrations, with a community repository (Splunkbase) for shared playbooks and connectors [8].

The strength is position in the Splunk ecosystem. Alerts from Splunk searches trigger SOAR playbooks with full access to the underlying log data, so enrichment steps can query the SIEM directly during execution. For organisations whose SIEM is Splunk, the integration is seamless.

**Strengths:** best-in-class Splunk integration; Python-based playbooks for maximum flexibility; mature platform.

**Weaknesses:** value is tied to the Splunk ecosystem; licensing costs add up fast; Cisco acquisition introduces uncertainty about product direction.

### 6.1.3 Microsoft Sentinel SOAR

Microsoft Sentinel is a cloud-native SIEM, but its SOAR capabilities are substantial. Automation rules and Logic Apps-based playbooks trigger on Sentinel analytics rule alerts. Logic Apps can call any Azure service, any Microsoft 365 API, and hundreds of third-party connectors [5].

In Microsoft-centric environments, this creates a unified SIEM + SOAR pipeline that sits directly behind Microsoft Defender XDR. XDR detections flow into Sentinel, which triggers Logic Apps playbooks for extended response. This is one of the cleanest examples of XDR and SOAR working together in a single vendor stack.

**Strengths:** cloud-native; no infrastructure to manage; direct XDR integration; Logic Apps connector library is massive.

**Weaknesses:** Logic Apps debugging is painful; cost model is per-action which gets expensive for high-volume playbooks; less flexibility than Python-based alternatives.

### 6.1.4 IBM QRadar SOAR

Formerly Resilient. Focuses on compliance-driven workflows. Maps response actions to regulatory frameworks—GDPR breach notification, HIPAA, PCI DSS—and tracks whether every required step has been completed. Particularly useful in regulated industries where the incident response process itself must meet specific documented standards [9].

**Strengths:** compliance mapping; structured task management; strong in regulated environments.

**Weaknesses:** integration library is smaller than XSOAR or Splunk SOAR; the interface feels dated; less flexible playbook logic.

## 6.2 Open-Source Options

**Shuffle** is an open-source SOAR with a visual workflow builder, a growing library of community playbooks, and webhook-based alert ingestion. Self-hosted, and suitable for small to mid-size teams that need basic orchestration without commercial licensing [13].

**TheHive + Cortex** combine case management (TheHive) with automated enrichment and response (Cortex). Cortex analysers query threat intelligence, sandbox files, and check reputations. Cortex responders execute containment actions. The pair approximates SOAR functionality with an emphasis on incident tracking [14].

**n8n** is a general-purpose workflow automation tool that can serve as a lightweight SOAR. It supports hundreds of integrations and a visual builder. Not purpose-built for security,

but capable of handling cloud-oriented security workflows—particularly when the main need is API orchestration rather than full case management.

## 7. Implementation in Practice

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### 7.1 Implementing XDR

XDR implementation follows a phased approach. Getting it wrong—connecting every data source at once and enabling full automation on day one—produces a flood of false positives that destroys analyst trust in the platform within weeks.

**Phase 1: Endpoints and identity.** Deploy the agent to endpoints and connect the identity provider. This covers the two most common attack paths: malware on endpoints and credential-based attacks. Run in detection-only mode (no automated response) for two to four weeks. Use this period to review alert quality, tune detection thresholds, and suppress known benign activity.

**Phase 2: Network and email.** Integrate firewall logs, proxy logs, DNS logs, and email platform data. This enables the correlation engine to link phishing-based initial access to endpoint activity and lateral movement. Adjust detection sensitivity based on Phase 1 false positive rates.

**Phase 3: Cloud.** Connect cloud provider audit logs, container runtime telemetry, and cloud identity events. This phase typically requires the most tuning because cloud environments generate high-volume, low-signal telemetry by default. Selective ingestion is important: forward high-value events (authentication, privilege escalation, security group changes, sensitive data access) and filter routine noise (health checks, read-only operations on non-sensitive resources).

**Phase 4: Automation.** With baselines established and false positives under control, begin enabling automated response for high-confidence detections. Document the criteria clearly: which detection types, at what confidence level, affecting which asset categories, trigger which response actions. Review every automated action within 24 hours for the first month.

### 7.2 Implementing SOAR

SOAR implementation centres on playbook development and integration engineering.

**Phase 1: Connect alert sources.** Integrate the SIEM, EDR, email gateway, and cloud security tools. Each integration needs testing: confirm alerts arrive in the expected format with the fields playbooks will need.

**Phase 2: Build high-value playbooks.** Start with the incident types that consume

the most analyst time and follow the most predictable steps. Phishing triage, malware alert enrichment, and suspicious login investigation are common starting points. Each playbook must be tested in a staging environment before production deployment—an untested playbook that blocks a legitimate user’s account is worse than no automation at all.

**Phase 3: Connect response tools.** Integrate the firewall, endpoint agent, identity provider, ticketing system, email platform, and cloud provider APIs for response actions. Each integration needs a dedicated service account with minimum required permissions. Over-privileged SOAR service accounts are a high-value target for attackers.

**Phase 4: Measure and expand.** Track mean time to respond, playbook success rate, false positive rate by playbook, and analyst time saved. Add playbooks for additional incident types. Build human-in-the-loop steps for high-impact actions.

### 7.3 Combined XDR + SOAR Implementation

In a combined architecture, the integration point is typically a webhook or API connection where XDR sends incident data to SOAR when an incident exceeds a severity threshold or matches a defined type. SOAR then runs extended response workflows: actions on tools outside the XDR ecosystem, case management with full audit trail, compliance-driven documentation, and notification chains.

The key design decision is what crosses the boundary. Sending every XDR alert to SOAR recreates the noise problem. Sending only correlated, high-confidence incidents keeps the SOAR pipeline clean and the playbooks effective.

## 8. Cloud-Specific Considerations

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Cloud environments change how both XDR and SOAR operate in ways that go beyond simply adding another data source.

### 8.1 Ephemeral Infrastructure

Containers run for minutes. Serverless functions execute for milliseconds. Auto-scaled instances appear and disappear based on load. A traditional endpoint agent that assumes a long-lived host does not work here.

XDR platforms have adapted in several ways. Some deploy sidecar containers that collect telemetry alongside application containers. Some use eBPF-based sensors attached at the kernel level, which survive individual container restarts. For Kubernetes environments, telemetry collection at the kubelet and API server level captures cluster-wide activity

regardless of individual pod lifetimes [15]. For serverless, telemetry comes from cloud provider APIs and function-level logging—there is no host to instrument.

For SOAR, ephemeral infrastructure means response actions must target the control plane. Isolating a compromised container means deleting the pod and tightening the network policy, not sending a command to an agent running inside it. Playbooks must handle the case where the targeted resource no longer exists by the time the action executes.

## 8.2 Multi-Cloud Complexity

An attack in a multi-cloud environment might start with a phishing email (Microsoft 365), compromise an identity (Entra ID), pivot to a cloud workload (AWS EC2), and exfiltrate data through a storage service in a different provider (GCP Cloud Storage). The full chain spans three vendors and two cloud platforms.

XDR’s cross-layer correlation is extremely valuable here, but only if the platform can ingest telemetry from every environment involved. In practice: Microsoft Defender XDR is deepest in Azure, CrowdStrike and Cortex XDR offer broader multi-cloud reach, and Elastic Security treats all sources equally because its ingestion pipeline is vendor-agnostic.

SOAR platforms are inherently multi-cloud capable because each provider is just another API integration. A single playbook can revoke an AWS IAM key, block an IP in Azure, and disable a GCP service account in sequence.

## 8.3 Cloud Provider Security Services

Each cloud provider offers native detection and monitoring services.

Table 2: Native cloud security services relevant to XDR and SOAR integration

| Provider     | Detection Services                                         | Key Telemetry Sources                                                        |
|--------------|------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>AWS</b>   | GuardDuty, Security Hub, Inspector, Macie, Detective       | CloudTrail, VPC Flow Logs, S3 access logs, EKS audit logs, Route 53 DNS logs |
| <b>Azure</b> | Defender for Cloud, Sentinel, Defender for Identity        | Activity Log, NSG flow logs, AKS audit logs, Entra ID sign-in logs           |
| <b>GCP</b>   | Security Command Centre, Chronicle, Event Threat Detection | Cloud Audit Logs, VPC Flow Logs, GKE audit logs, Access Transparency logs    |

Both XDR and SOAR need to integrate with these services: XDR ingests their telemetry for correlation, SOAR calls their APIs for response. A well-designed architecture uses

cloud-native services as data sources feeding into XDR, with SOAR handling orchestration for actions that span multiple providers.

## 8.4 Identity as the Primary Attack Surface

Cloud environments have no traditional network perimeter. Identity—credentials, API keys, service account tokens, OAuth grants—is the attack surface. Lateral movement happens through role assumption and privilege escalation, not network pivoting [19].

Identity telemetry is therefore the most important data source for both XDR and SOAR in cloud environments. XDR needs real-time sign-in events, privilege changes, and token issuance to detect credential-based attacks. SOAR needs the ability to revoke tokens, rotate secrets, disable accounts, and adjust IAM policies as response actions.

This is also where the gap between XDR and SOAR is clearest. XDR can detect an anomalous role assumption in real time and kill the session. SOAR can run a broader workflow afterward: revoke all sessions for the compromised identity, audit every resource it accessed, rotate related secrets in the vault, check for persistence mechanisms (new access keys, new roles, new service accounts), notify the affected team, and create a compliance record.

## 8.5 Data Volume and Cost

Cloud environments generate large telemetry volumes. A moderately sized AWS deployment can produce hundreds of gigabytes of CloudTrail and VPC Flow Log data per day. Ingesting all of it into an XDR platform or SIEM at per-GB pricing is expensive.

Effective cloud monitoring requires selective ingestion. High-value events— authentication, privilege escalation, security group modifications, data access on sensitive resources— should always be ingested. High-volume, low-value events—routine API calls, health checks, read-only operations on non-sensitive resources—should be filtered at the source or sent to cold storage for later retrieval if needed.

Both XDR and SOAR configurations need to account for this. XDR platforms with flat-rate pricing (CrowdStrike Falcon, for example) are more predictable for high-volume cloud environments than those that charge per GB ingested.

## 9. Combined Architectures for Advanced Monitoring

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In a mature cloud security operation, XDR and SOAR are not alternatives. They are layers. The reference architecture looks like this:

**Layer 1: Telemetry.** Cloud provider audit logs, endpoint agents, network sensors,

identity provider events, and email data all flow into the XDR platform. Cloud-native detection services (GuardDuty, Defender for Cloud, SCC) also feed alerts into XDR for correlation.

**Layer 2: Detection and correlation.** The XDR platform normalises, correlates, and applies its detection engine to the combined telemetry. Raw events become correlated incidents with timelines, severity scores, ATT&CK mappings, and affected asset inventories. First-line automated responses (endpoint isolation, IP blocking, account suspension) execute within the XDR platform for high-confidence threats.

**Layer 3: Orchestration and extended response.** Correlated incidents above a severity threshold are forwarded to SOAR. SOAR playbooks execute extended workflows: cross-cloud response actions, third-party tool actions, case management, compliance documentation, notification chains, and evidence preservation.

**Layer 4: Feedback.** Post-incident reviews feed improvements back into all layers. New detection rules for the XDR platform. Updated playbooks for SOAR. Adjusted ingestion filters for cloud telemetry. Coverage gap analysis against ATT&CK.

This architecture gives the organisation the detection quality of XDR (reducing alert noise before it reaches SOAR), the orchestration breadth of SOAR (reaching tools beyond XDR’s ecosystem), and a complete audit trail for compliance.

The practical challenge is maintaining it. Each integration needs monitoring. Each playbook needs periodic review. Each detection rule needs tuning as the environment changes. Without ongoing operational investment, even the best architecture degrades.

## 10. Conclusion

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XDR and SOAR are different tools for different problems. XDR consolidates detection. It replaces disconnected security tools with a single platform that collects telemetry from every layer, correlates events across sources, and responds to threats it finds. It directly reduces alert noise and gives analysts complete incident timelines instead of fragments.

SOAR consolidates response. It connects existing tools through APIs, automates multi-step workflows that analysts previously ran by hand, and provides the case management and audit trail that security operations and compliance demand. It does not generate detections—it assumes someone else has that covered.

In cloud environments, both face specific challenges. Ephemeral workloads resist traditional agent-based monitoring. Multi-cloud deployments spread telemetry across providers with different APIs, schemas, and security services. Identity replaces the network as the primary attack surface. Data volumes drive ingestion costs that can grow unpredictably.

The right choice depends on where the bottleneck is. If the problem is detection quality—too many disconnected tools, too many uncorrelated alerts, blind spots between layers—XDR addresses that directly. If the problem is response speed and consistency—solid detection but slow, manual, inconsistent containment workflows—SOAR addresses that directly. If both are bottlenecks, deploy both: XDR for detection and first-line response, SOAR for extended orchestration and case management.

None of this works without operational commitment. The platform that gets deployed but not tuned, maintained, and staffed ends up as another underused tool generating noise. The technology is the smaller part of the problem. The people and processes around it are what determine whether it delivers.

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